

Multi-Dimensional Coverage Profiles: A Lexical, Affective, and Moral Comparison of How Five U.S. News Outlets Cover Five Contested Topics

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1. Abstract

Existing media-bias rating systems characterize news outlets primarily along a partisan left–right axis, an approach that can flatten editorial behavior into a single label and leave readers without the detail needed to understand how outlets actually frame the news. This paper builds multi-dimensional coverage profiles for five news outlets, CNN, BBC, Associated Press, Fox News, and NewsMax, across five contested topics: artificial intelligence, climate, the economy, immigration, and Iran. The analysis combines six complementary methods: TF-IDF lexical signatures, VADER sentiment, NRC discrete-emotion scoring, Moral Foundations Dictionary scoring, per-topic non-negative matrix factorization, and spaCy named-entity recognition. These methods are applied to a corpus of 2,334 full-text articles collected via MediaCloud and scraped with trafilatura. The resulting profiles surface patterns single-axis bias ratings cannot capture: Fox News intensifies fear-coded vocabulary in headlines while CNN leads on anger-coded vocabulary; NewsMax is an outlier on fairness-coded moral language; BBC sits structurally outside the U.S. political register entirely; and outlets across the political spectrum foreground entities representing the opposite of their stated ideological lean. A case study on immigration coverage demonstrates that these per-outlet patterns sharpen rather than dissolve under topic constraint. An accompanying interactive web application allows readers to browse the profile components for any outlet–topic combination, serving as a prototype for a reader-facing media-literacy tool.

2. Introduction

Americans are consuming more news than ever and trusting less of it. Pew Research finds that half of U.S. adults report that Americans on the whole consume more news than they did a decade ago, (Pew Research Center, 2026) while confidence in national and local news organizations has declined across age cohorts and the political spectrum. (Pew Research Center, 2025) At the same time, the way Americans encounter news has shifted. Where readers once approached a small set of familiar outlets directly, social feeds now deliver a broader and more shifting set of sources. Many of these sources are unfamiliar enough that readers have little basis to evaluate them, and are easy to dismiss if they appear to come from ‘the other side’. Meanwhile, the algorithmically curated social feeds can guide readers to news content along familiar grooves, so that a broader pool of outlets is filtered through narrower lenses not directly chosen by, or ever explained to, the reader. The net effect is that Americans face more news, from more sources, with less of their own choosing and less context about who is producing it. They have no straightforward way to understand and compare how these sources are framing the news and how story coverage compares with other sources.

The purpose of this paper is to address this comparison problem. A reader who wants to know how an unfamiliar news outlet, or their preferred news source, frames a topic differently from other outlets has few practical choices. The most visible tools, such as AllSides, Ad Fontes Media, Media Bias/Fact Check, all chart news outlets on a simple left/right axis. These scores are useful as orientation, but they hide

information that may be of actual use to readers as they try to understand how news is framed. This includes which entities an outlet foregrounds, which moral vocabularies it invokes, and where its headlines diverge from its body text. A single bias score will show the reader which side an outlet sits, but will not answer “how”?

We build a multi-method, multi-dimensional coverage profile tool that aims to answer the latter. The tool can be previewed on huggingface.co at the following link: <https://huggingface.co/spaces/Jack-O-Wood/news-outlet-profiles>. Rather than collapse each outlet to a single score, outlets are measured here along six complementary dimensions: distinctive vocabulary, sentiment, discrete emotion, moral language, sub-theme emphasis, and named-entity prominence. The goal is explicitly not to assign a bias label to any outlet, but to surface patterns a reader could review and interpret on their own terms. The resulting profiles reveal divergences in headline emotion, moral vocabulary, and entity selection that single-axis bias scoring cannot capture.

3. Data

The dataset for this research was compiled via MediaCloud API for article discovery and trafilatura for full text extraction. News outlets were selected to cover the political spectrum and maximize data availability, as many widely read news outlets restrict API access or sit behind paywalls, and therefore not accessible by the API compiling and article scraping methods. The final selection includes: CNN (left-leaning), BBC (international-center/center-left), AP (center wire service), Fox News (right-leaning), NewsMax (right-leaning). Outlet placement on the political spectrum follows the Ad Fontes Media Bias Chart, which rates individual articles along two axes, political bias (x), and reliability (y). (Ad Fontes Media, 2026) The Media Bias Chart uses a panel of analysts spanning the political spectrum, then aggregates article level scores into outlet level placements (*Figure 1*)

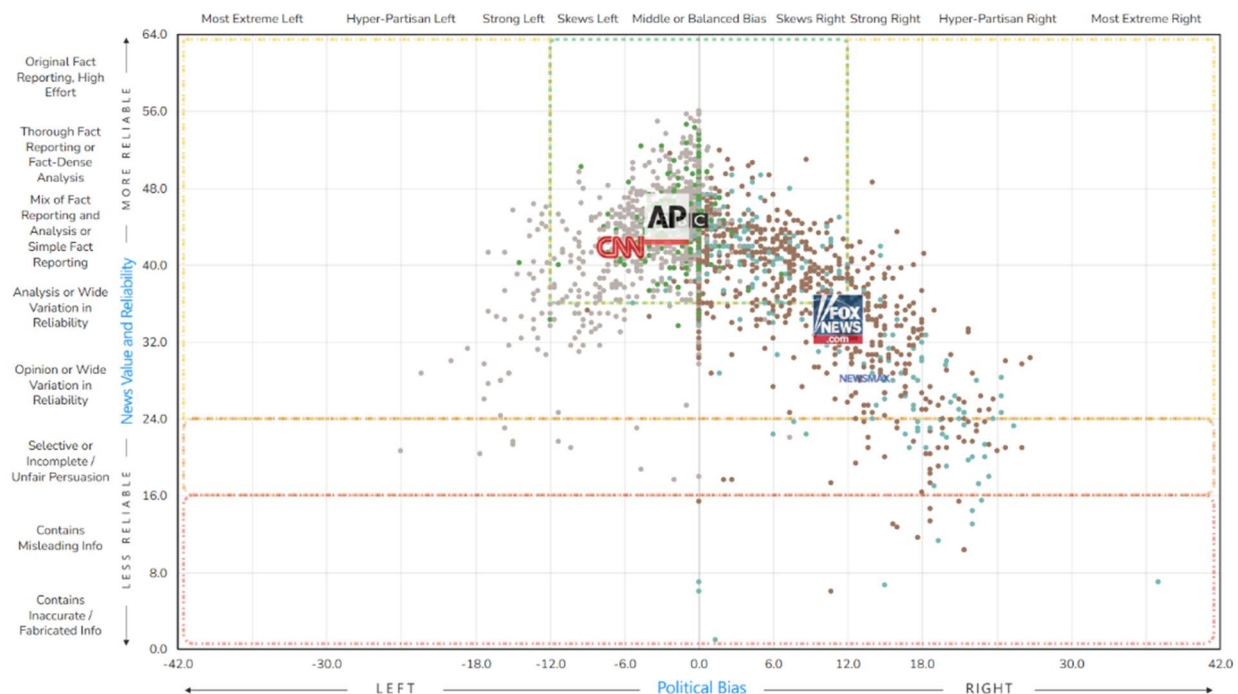


Figure 1, Ad Fontes Media Bias Chart with selected outlets marked

Initially, the project was planned around event-matched sampling but found that manual event matching for same day reporting across all outlets was too difficult, and would not generate a large enough corpus for meaningful analysis. Topic-based sampling trades direct event comparability for the within-outlet sample sizes needed to characterize outlets in aggregate.

The article base is comprised of topic categories; AI, Iran, Climate, Immigration, and Economy. These topics were chosen to cover a mix of policy domains likely to see diverging framing. For each of these topics, a list of keywords was used to search for articles that correspond with each category. For example, the climate topic included: "climate change," "global warming," "carbon emissions," "climate crisis". The articles were also limited to the date range of 1/1/26 to 4/22/26. The MediaCloud API was queried for each (keyword $\times$ outlet) pair over the window of January 1 to April 22, 2026, yielding 19,554 metadata records after cross-keyword URL deduplication.

From this pool of article metadata, a balanced sample was taken of up to 100 articles per topic per outlet, producing roughly 3,000 candidate URLs. The full text of each article was then scraped using trafiletura. After scraping, the dataset was then cleaned, removing articles with under 200 words, filtering for English only articles, and removing duplicate articles. This final article count after cleaning was 2,334 articles. Language filtering was applied after rather than before sampling; in practice this affected fewer than 4% of scraped articles and left the corpus well-balanced across both outlet and topic margins. Full cell counts appear in Table 1.

Table 1. Article counts by topic $\times$ outlet in the final corpus (N = 2,334)

Topic	AP	BBC	CNN	Fox News	NewsMax	Total
AI	86	73	99	100	100	458
Climate	99	99	99	79	45	421
Economy	100	98	99	100	100	497
Immigration	100	99	100	100	100	499
Iran	99	60	100	100	100	459
Total	484	429	497	479	445	2,334

4. Methods

4.1 Preprocessing

Headlines and article bodies were analyzed as separate text fields throughout. Three text representations were produced and used by different downstream methods. First, a raw representation was used that preserves the original casing, punctuation, and intensifiers. This is required by VADER, which weights based on capitalization and exclamation, and by spaCy's named entity recognizer, which depends on casing. Second, a normalized representation lowercases the text, tokenizes using NLTK, drops non-alphabetic tokens, and removes NLTK's English stopwords list. Third, a topic-clean representation extends the normalized form with the below mentioned per-document blocklist. This third representation is the input to all the TF-IDF and NMF analysis.

The topic-clean representation addresses a problem specific to topic-conditioned corpora: when articles are sampled by keyword, the keyword and its near-synonyms dominate TF-IDF and NMF results across every outlet, swamping the within-topic differences that the analysis is trying to surface. Therefore a three-layer stopwords list was applied at document tokenization. First, a manual base list of the topic label tokens was assembled, such as “climate”, for the climate topic or “Iran” and “Iranian”, for the Iran topic. Next, an auto-detected list per topic of cross outlet vocabulary was added to the stopwords list. For each topic, tokens were removed that fit the following three criteria: (i) appear in at least one article from every outlet within the topic, (ii) occur at least 12 times in total across the topic, and (iii) rank in the top 40 by total frequency. Finally, a list of global news-related words were added to the stopwords list, such as “watch”, “video”, and “said”. The per-document blocklist is then the union of these three layers plus the tokenized form of the specific keyword that retrieved each article, ensuring that an article pulled by "carbon emissions" does not register "carbon" or "emissions" as outlet-distinctive vocabulary. The final topic-clean blocklist is applied only to lexical methods (TF-IDF, NMF). Sentiment (VADER), discrete-emotion (NRC), moral-foundations (MFD), and entity-recognition (spaCy) methods operate either on raw text or on lowercase tokens without the blocklist, because in those analyses the topic-specific terms themselves may carry the affective, moral, or entity-level signal of interest.

4.2 Lexical distinctiveness — TF-IDF

Term Frequency–Inverse Document Frequency (TF-IDF) was used to surface vocabulary that distinguishes one outlet's coverage from others'. A term scores high when it appears frequently in one outlet's articles but rarely across the rest of the corpus. In this context, this shows outlet-distinctive framing vocabulary. All TF-IDF analyses consume the topic-clean text representation, ensuring that topic-label and keyword vocabulary do not dominate the rankings.

TF-IDF was used to surface vocabulary that distinguishes one outlet's coverage from others'. Each outlet's articles were concatenated across all five topics into a single per-outlet pseudo-document and ran scikit-learn's `TfidfVectorizer` over the resulting five-document corpus. Under this aggregation the IDF term penalizes vocabulary shared across outlets, leaving terms that characterize one outlet's overall coverage. The same procedure was applied separately to headlines. Raw counts from `CountVectorizer` were retained alongside TF-IDF scores so that both frequency and distinctiveness could be reported. It is important to note that in this case TF-IDF does not necessarily surface vocabulary that is important within any one outlet. Terms used heavily but in roughly equal proportion by all outlets, such as the names of major political figures, are deliberately suppressed by IDF weighting. This is the appropriate behavior for surfacing framing differences but means TF-IDF rankings are not a stand-in for the most mentioned topics.

4.3 Sentiment — VADER (body, headline, divergence)

Sentiment was measured at the article level using VADER (ValenceAware Dictionary and Sentiment Reasoner; Hutto & Gilbert, 2014) implemented in NLTK's `SentimentIntensityAnalyzer`. VADER returns four scores for a given text, positive, neutral, and negative proportions plus a normalized compound score ranging from -1 to 1. It was applied to the raw text representation, which preserves the casing, punctuation, and intensifiers that the VADER lexicon weights. The compound score is the primary metric used throughout.

Each article's body and headline were scored independently. Article-level scores were then aggregated to two levels: a per-outlet mean compound across all topics, and a per-outlet and topic mean compound for finer-grained breakdowns. Both aggregations were computed separately for bodies and headlines. In addition, a divergence variable was calculated by subtracting the body compound score from the headline compound score.

Under this convention, a negative divergence indicates that the headline is more negative than the body, while a positive divergence indicates that the headline is more positive than the body. Divergence was aggregated to outlet-level means and to outlet $\times$ topic means; the per-outlet distribution of divergence scores was also examined to identify articles with unusually large headline–body gaps.

4.4 Sub-theme extraction — per-topic NMF

To surface latent sub-themes within each topic, non-negative matrix factorization (NMF) was applied to each of the five topic-conditioned corpora. Running NMF per topic rather than globally allowed sub-themes to emerge conditional on the topics. For example, “fiscal policy” and “monetary policy” were distinct sub themes within Economy. For each topic, articles were vectorized over the topic-clean text representation with unigrams and bigrams. NMF was then fit on the resulting matrix, yielding 5 sub-themes per topic and a document component matrix H giving each articles loading on each sub-theme. Sub-themes were characterized by their top 10 terms from H . Outlets were scored on each sub-theme by averaging the per-document W weights of their articles within the topic, then normalizing across the five sub-themes so that each outlet's emphasis sums to one. This produces an outlet $\times$ sub-theme proportions matrix that is directly comparable across outlets for any single topic. After extraction, human-readable labels were manually assigned to each sub-theme based on the top terms. These labels are thus interpretive aids for reporting and not algorithmic results.

4.5 Moral foundations — Moral Foundations Dictionary

Moral Foundations Theory (MFT), introduced by Haidt and Joseph (2004) and elaborated by Graham, Haidt, and Nosek (2009), proposes that human moral judgment rests on a small set of innate intuitive foundations rather than a single underlying axis of right and wrong. The framework identifies five foundations:

- a. Care/Harm (concern for the suffering and protection of others)
- b. Fairness/Cheating (proportionality, equality, and justice)
- c. Loyalty/Ingroup (allegiance to one's group, nation, or community)
- d. Authority/Subversion (deference to legitimate hierarchy, tradition, and order)
- e. Purity/Sanctity (cleanliness, religiosity, and bodily integrity)

A body of empirical work has shown that political liberals tend to weight Care and Fairness most heavily while political conservatives draw on all five more evenly. The framework is therefore well-suited to comparative media analysis. An outlet's relative emphasis on each foundation offers a window into the moral register through which it frames events.

To measure the moral foundations in the articles, the Moral Foundations Dictionary was used to define word lists for each of the five foundations. Following the dictionaries conventions, each list contained both the virtue and vice terms. This means that the moral foundation scores measure engagement with one

of the five foundations positively and negatively combined, regardless of the evaluative direction. Therefore an article with many “care” words and another with many “harm” words would both score high on the Care/Harm foundation.

For each article, the raw body text was tokenized by lowercasing, replacing non-alphabetic characters with spaces, and splitting on whitespace, yielding a token stream that was then matched against each foundation's word list. Dictionary terms support prefix wildcards (e.g., 'care*' matches "care," "caring," and "careful"), implemented as a starts-with comparison against each token. Per-article match counts were converted into a density score by dividing the number of dictionary hits in the article by its total token count, controlling for length so that longer articles do not mechanically accumulate more matches. The moral foundation density scores were aggregated on the outlet level and also per outlet-topic level.

4.6 Discrete emotion — NRC Emotion Lexicon (with density correction — note why)

To complement the moral vocabulary analysis, the articles were scored against the NRC Emotion Lexicon (EmoLex; Mohammad & Turney, 2013), a word list whose entries are each crowd-annotated for association with eight discrete emotions: anger, fear, disgust, sadness, joy, anticipation, surprise, and trust. Unlike VADER's compound score, which collapses affect onto a single positive-negative axis, the NRC emotion lexicon differentiates between more detailed emotions. This allows for a more granular understanding of what emotions certain outlets or topics are evoking in the articles.

An emotion density was computed for each article for each of the eight emotions, similar to the MFT density scoring, but scaled to per-100 words. The same procedure was applied to the body text and headlines, using raw text as tokenization is handled internally by the nrclex library. Density was preferred over raw count as article size varied between outlets and a raw count could boost longer article's scores.

In addition to the eight emotion density scores, a composite “outrage” score was calculated as the sum of the anger, fear, and disgust densities, capturing high-arousal negative emotions in a single dimension. In order to understand how this “outrage” emotion language is used in headlines compared to in the body of the article, an “outrage gap” dimension was created, where a positive value indicates that an article's headline carries a higher density of high-arousal negative emotion than its body. This is a more affective analogue of the VADER divergence score in Section 4.3. Densities and outrage gaps were aggregated to outlet means, topic means, and by outlet-topic means.

4.7 Named entity recognition — spaCy

To examine which people, organizations, and places each outlet foregrounds, named entities were extracted from article bodies using the spaCy package. Attention was restricted to four entity types relevant to media framing: PERSON (individual people), ORG (organizations and agencies), GPE (geopolitical entities, countries, states, cities), and NORP (nationalities, religious groups, and political groups). The raw NER was then cleaned as many entities appeared in multiple forms across the corpus, such as “Trump”, “President Trump”, “Donald Trump” all being the same entity. A manual normalization map was built to collapse these forms into a single variant. The map was constructed iteratively by inspecting the most frequent entities and folding in obvious variants. Second, many false positives, such as the outlets themselves or reporters names were removed. Scraping artifacts from navigation menus or website footers were also removed. After normalization and filtering, per-article entity lists were exploded

into a long-format table indexed by outlet, topic, entity, and entity type, supporting per-outlet rankings of the most frequently mentioned entities of each type and per-outlet-topic breakdowns.

5. Results

The cross-outlet comparison follows five threads: how outlets differ in their headlines and bodies, the emotional and moral register of their coverage, lexical signatures, the topic-level framings they bring to a single news cycle, and the entities they foreground. Each thread draws on more than one of the analytic dimensions described in the methodology, as the paper's central claim is that no single dimension surfaces a meaningful outlet profile on its own.

5.1 Headline Sensationalism

According to the VADER sentiment analysis, all outlets have negative sentiment headlines and all outlets' headlines are more negative than their article bodies. (Figure 2) CNN on average has the most negative article bodies, while BBC has noticeably more positive article bodies. This is consistent with BBC's NMF profile, where it loads more heavily than peer outlets on the AI sub-theme labeled 'Lifestyle & culture verticals'. Due to the positivity in the article bodies, BBC shows the largest headline-body divergence (mean compound shift of -0.29). Fox News has the most negative headlines with slightly positive bodies, making for the second largest sentiment gap. CNN on the other hand has a smaller sentiment gap, largely due to having the most negative article bodies.

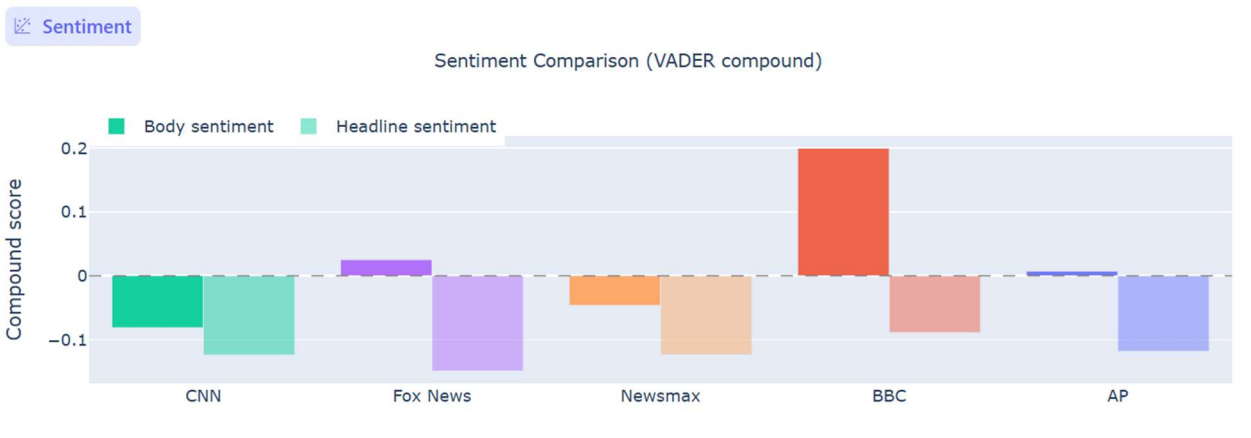


Figure 2, VADER sentiment for all outlets' Headlines and Bodies

All news outlets are using more emotional language in their headlines than in the bodies of the articles, across all emotions. The NRC outrage gap, which measures the difference per-100-words in anger, fear, and disgust vocabulary between article bodies and headlines, surfaces a different outlier. Fox News stands out sharply (+6.75), with a gap roughly forty percent larger than the next-highest outlet. NewsMax, CNN, and AP cluster well below Fox at moderate outrage gaps, while BBC again sits at the low end, consistent with its overall positive register. (Figure 3) This pattern aligns with the VADER sentiment results, where Fox News also produces the most negative headlines in absolute terms. Both measures point to Fox News's headlines as unusually intensified relative to its body content

However, the two sensationalism measures, VADER sentiment gap and NRC outrage gap point to different outliers, BBC by VADER divergence, Fox News by NRC outrage. BBC scores highest on

VADER divergence but lowest on the NRC outrage gap, suggesting that BBC's sentiment divergence is driven less by sensationalist headline framing than by the unusually positive baseline of its lifestyle-oriented body content. By the outrage-gap measure, which is less sensitive to body-content positivity, BBC sits at the opposite end of the corpus from Fox News

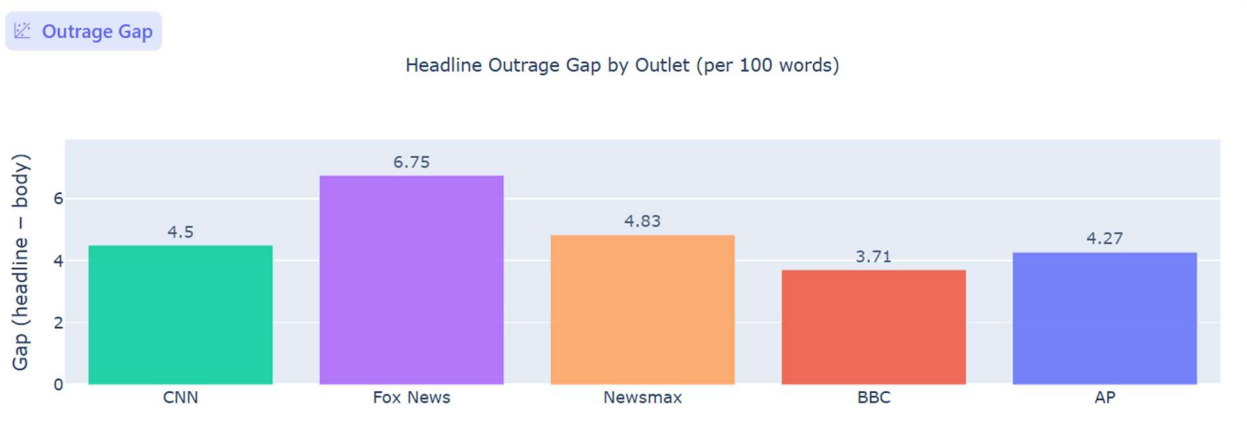


Figure 3, Outrage Gap across 5 outlets

5.2 Emotional and Moral Register

Fox News uses the most emotional language in its headlines across nearly every emotion measured: it ranks highest on fear, disgust, sadness, surprise, and trust, and second-highest on anger and joy. Its emotional intensity is broad rather than narrow. CNN's pattern is the inverse. CNN leads on anger but sits at the bottom of the corpus on joy, anticipation, and trust, suggesting that anger functions as a distinctive register rather than as part of a broader emotional palette. This difference can be seen most clearly in immigration coverage, where CNN's anger density reaches 7.71 words per 100 tokens (next closest Fox News at 5.78), while Fox News's fear density reaches 7.63 (next closest NewsMax at 5.63). BBC emerges as a positive outlier again, with the highest joy and anticipation scores in the corpus and the lowest density of disgust and anger. (Figure 4)

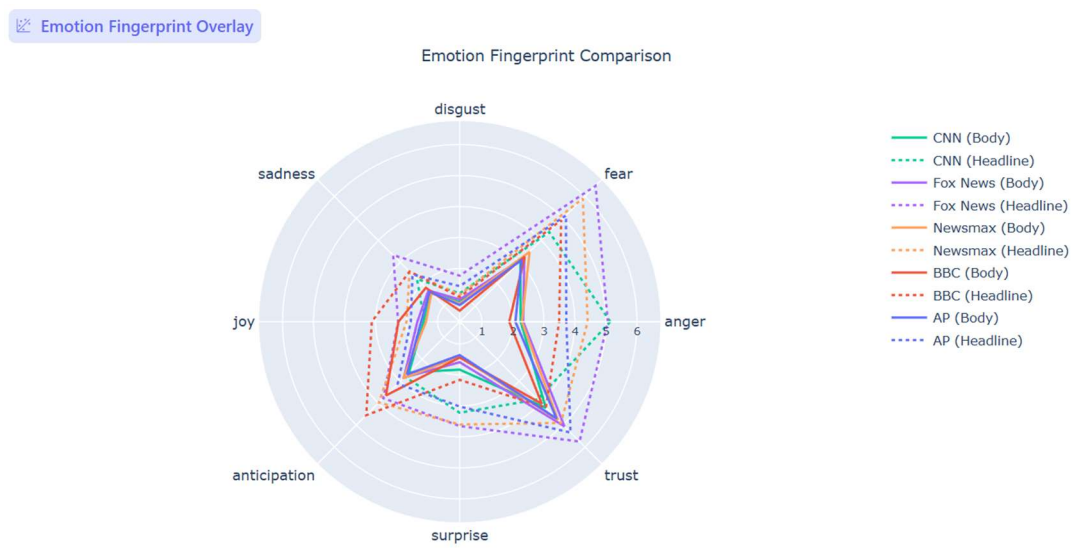


Figure 4, Radar chart of NRC emotional densities per outlet, split headline and body

Comparing body to headline emotion density reveals a notable editorial divergence. Across all eight NRC emotions, outlets are remarkably similar in body text: total emotional density ranges only from 17.9 (CNN, AP) to 19.0 (BBC). In headlines, the same total ranges from 24.2 (CNN) to 32.1 (Fox News). The cross-outlet emotional differentiation is therefore greater at the headline level. Fox News exhibits the largest amplification, as its headline emotional density is 74% higher than its body emotional density. BBC and CNN on the other hand amplify least (37% and 35%). This pattern mirrors the lexical finding in Section 5.3, where Fox News's distinctive editorial vocabulary was shown to be concentrated in headline framing rather than woven through body text. Two methods independently locate Fox News's distinctiveness at the headline level.

Two other outlet patterns hold across both body and headline. CNN's elevated anger and depressed joy appear in body text as well as headlines, suggesting that anger functions as a consistent editorial register rather than as a headline-only device. BBC's high joy and anticipation also extend through body text, reinforcing the reading that BBC's positive register is structural to its coverage rather than an artifact of headline-construction.

Moral Foundations analysis revealed sharper outlet-level divergences than overall sentiment (*Figure 5*). Care/Harm and Authority dominated across all five outlets, while Purity scores were uniformly low (~1 per 1,000 words), suggesting that sanctity-based moral language does not meaningfully distinguish contemporary U.S. news coverage. Fox News led on Authority (8.5 per 1,000 words, vs. CNN's 6.8 and BBC's 5.5), consistent with right-leaning outlets' canonical emphasis on the binding foundations in MFT literature. NewsMax was the most heavily moralizing outlet overall, leading on Care/Harm and Fairness, with Loyalty and Authority close to Fox's. NewsMax was a large outlier on Fairness, scoring 2.5 against the corpus mean of approximately 1.2. This margin suggests Newsmax's coverage emphasizes fairness/justice/cheating-coded vocabulary substantially more than peer outlets. BBC scored the lowest across four of the five foundations, reflecting a consistently less moralizing register. (*Figure 5*)

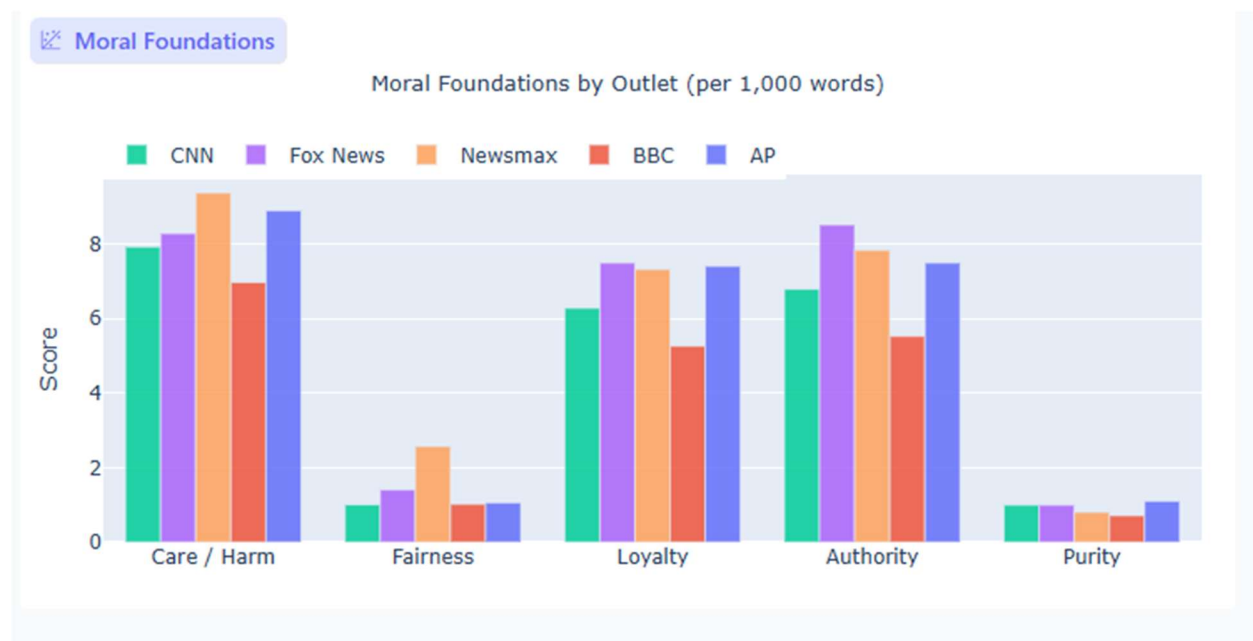


Figure 5, grouped bar chart by Moral Foundation word density per outlet

5.3 Lexical Signatures

TF-IDF analysis of body text surfaces vocabulary that distinguishes each outlet from the rest of the corpus, and the patterns add lexical grounding to the affective and moral findings reported above. (*Figure 6*) Fox News's distinctive signature is concentrated in headlines rather than bodies. Its top headline terms, “gop”, “dem”, “dhs”, “warns”, “liberal”, “republicans”, “claims”, “pushes”, and “calls” combine partisan shorthand with alarm-coded verbs distinctive from other outlets top vocabulary. These terms rarely surface in other outlet's top headline TF-IDF rankings, indicating that Fox News's headline framing language is the most lexically distinctive in the corpus. Fox's body vocabulary, by contrast, runs closer to the corpus mean, suggesting that the partisan-alarm framing is concentrated in headline construction rather than woven through the article text itself.

NewsMax's distinctive vocabulary spans two related coverage emphases: national security and enforcement (“war”, “military”, “dhs”, “tsa”, “Mideast”, “strait”) and economic markets and prices (“oil”, “prices”, “energy”, “stocks”, “fed”, “deal”, “funding”). Together these suggest a “threats to America” framing that pairs geopolitical risk with economic anxiety. This pattern is sustained across both body and headline rankings, indicating that NewsMax's threat-and-markets framing is woven through the article text rather than concentrated in attention-grabbing headlines as with Fox News. One distinct term, “war”, overlaps directly with the MFD Care/Harm dictionary, which aligns with NewsMax's higher Care/Harm density score.

CNN's distinctive terms emphasize Trump-administration coverage and reactive analysis. In headlines, “trumps”, “iran”, “analysis”, “reacts”, “updates”, and “know”, cluster around explainer-style and reactive framing. In bodies, the prominence of “trumps”, “house”, and “iran”, points to attribution-heavy political reporting on the Trump administration. The reactive framing is consistent with CNN's role as a left-leaning outlet covering a right-leaning administration.

AP and BBC do not provide as clear signals through their lexical patterns. The AP top terms most closely follows the corpus average, which may be expected from the news wire service. The only outlet-unique body term is “israel”, which suggests slightly more Middle-East-focused coverage but is not a strong-enough signal to characterize the outlet on its own. BBC on the other hand is only notably distinct in its dominant UK regional markers in both bodies and headlines, such as “uk”, “Scotland”, “wales”, and “Yorkshire”. Considered together, the lexical signatures align with several of the affective and moral patterns reported in 5.1 and 5.2 without fully explaining them. Fox News's partisan-alarm headline vocabulary parallels its high outrage gap, NewsMax's security-and-economic threat vocabulary is consistent with its higher Care/Harm density, and CNN's reactive headline framing aligns with its outsized anger-density score. AP and BBC do not produce strong lexical signatures that map onto the affective or moral patterns.

	outlet	top_body_terms	top_headline_terms
0	AP	people, state, president, last, first, trumps, security, government, according, military, since, many, israel, time, like	administration, energy, war, plans, world, big, say, iran, first, oil, prices, least, highlights, photos, top
1	BBC	people, uk, government, last, world, time, told, first, scotland, home, energy, since, wales, like, make	uk, council, england, iran, scottish, west, yorkshire, war, reform, south, six, ni, centre, headlines, prices
2	CNN	people, president, told, last, according, trumps, house, time, like, iran, first, war, security, state, even	trumps, iran, analysis, war, updates, reacts, day, say, even, oil, know, february, administration, democratic, change
3	Fox News	people, president, state, like, security, told, house, time, first, democrats, american, states, administration, data, energy	gop, dem, dhs, trumps, house, warns, liberal, republicans, claims, amid, iran, pushes, top, sen, calls
4	Newsmax	prices, iran, people, security, energy, president, military, war, including, strait, oil, administration, global, may, government	iran, dhs, war, energy, oil, fed, prices, tsa, gop, deal, funding, amid, mideast, stocks, may

Figure 6, Aggregate per-outlet TF-IDF signatures

5.4 Entity Asymmetry

News outlets foreground political entities associated with the opposite of their stated ideological lean. Right-leaning outlets mention left-coded figures more prominently. For example, Fox News is the only outlet with Barack Obama in its top 10 PERSON entities and Fox and NewsMax both feature California in their top 10 GPE entities, where no left-leaning or center outlet does. Left-leaning CNN, in turn, devotes substantially more attention to right-coded administration figures than either right-leaning outlet does: CNN mentions Donald Trump 1,354 times, nearly twice as often as Fox News (765) and more than twice as often as NewsMax (612). Marco Rubio is mentioned by CNN (81) and AP (76) substantially more than by Fox (28) or NewsMax (35), and Pete Hegseth, formerly a Fox News host and now Trump's Defense Secretary, appears in CNN's top 10 PERSONs but in neither Fox nor NewsMax's. (Figure 7)

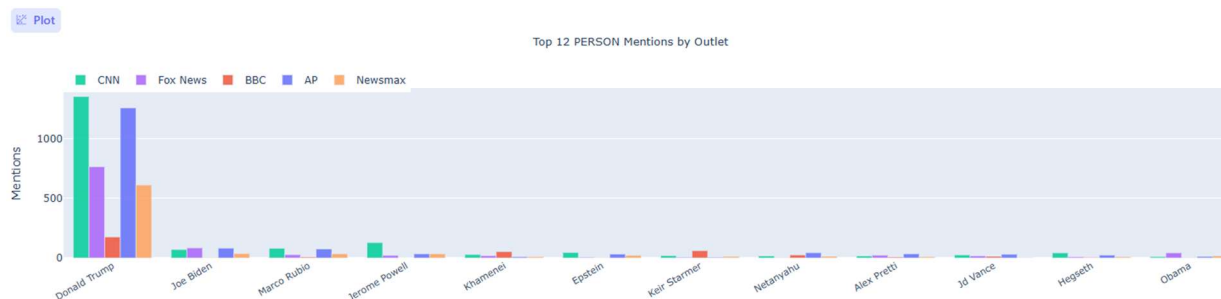


Figure 7, Top PERSON entity mentions by outlet

Organizational entity choices reveal distinct editorial priorities. Fox News uniquely features DEI (Diversity, Equity, Inclusion) in its top 10 ORGs as a culture-war framing marker. Meanwhile, NewsMax leans heavily on enforcement and security agencies (DHS, TSA, Pentagon) in line with the TFIDF findings. BBC's entity profile is built around its more geographically relevant UK and EU institutions

(Labour, NHS, SNP, EU) rather than U.S. political bodies. These outlet-specific entity choices reinforce the broader pattern that no single editorial dimension fully characterizes any outlet, but considered together they form a coherent profile.

AP's entity profile is the most internationally focused of the corpus with five world leaders (Macron, Putin, Maduro, Netanyahu, Khamenei via NORP) appearing in its top PERSON entities. Additionally, AP's top entities include the UN, Hezbollah, and a wide international GPE distribution (Ukraine, Lebanon, Russia, China, Greenland). AP is also the only outlet whose partisan NORP coverage approaches balance, with Democrats at 252 vs. Republicans at 195 (ratio 1.29). Other outlets range from 1.4 to 1.9 ratios in Democrat – Republican mentions. These results provide a quantitative anchor for the wire-service neutrality earlier observed lexically.

6. Use Case: Cross-Outlet Framing on a Single Topic

The analysis above supports two complementary lenses. The first is the per-outlet view which characterizes each outlet across all analytic dimensions and asks what distinguishes it from its peers. This is the lens Section 5 develops, and the lens the interactive explorer is built around.

The second lens is the inverse, holding a single news cycle constant and comparing outlets across dimensions, asking how the same underlying events are framed differently across the political spectrum. This section demonstrates that lens with immigration coverage. Immigration was chosen as it is the topic on which the outlets diverge most sharply in the data lexically, emotionally, and politically. The result is five concretely different framings of the same set of events, each consistent with the outlet-level characterizations from Section 5 but visible here at the granularity of a single topic.

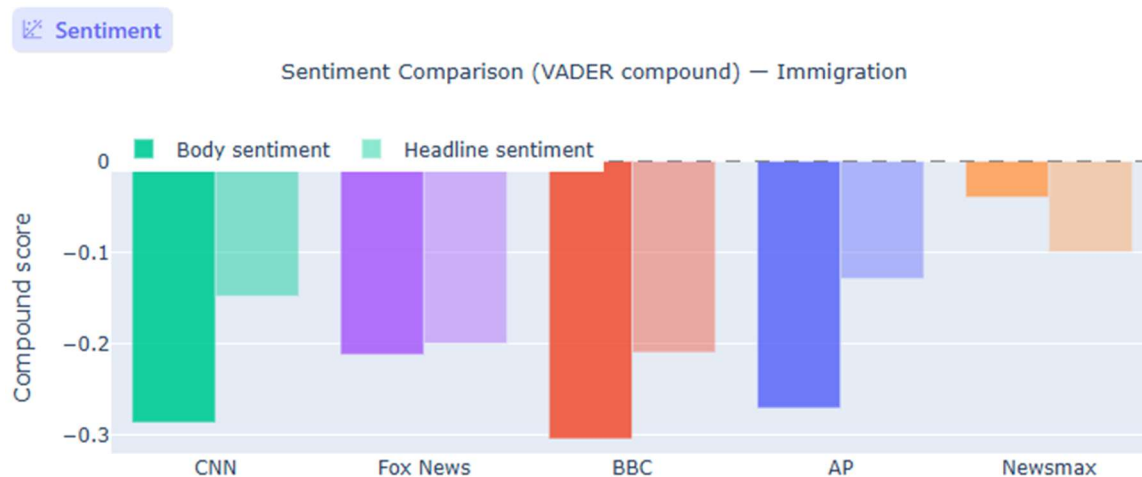


Figure 8, VADER sentiment for all outlets' Headlines and Bodies filter to Immigration topic articles

Sentiment shifts substantially on Immigration. Body sentiment is negative across every outlet on this topic. This pattern contrasts with the corpus aggregate, where BBC was substantially positive and Fox News and AP were near-neutral. The headline-body relationship also inverts. Where the broader corpus shows headlines more negative than bodies (Section 5.1), within the Immigration topic, four of the five outlets show bodies more negative than headlines, suggesting that the violence- and enforcement-coded vocabulary defining Immigration coverage concentrates in body text rather than being amplified at the

headline level. NewsMax is the lone exception. Its body sentiment is nearly neutral (-0.04) and its headlines are slightly more negative (-0.10). This is consistent with NewsMax's focus on the DHS and TSA funding issue as opposed to broader immigration issues. This focus is visible in its distinctive vocabulary and entity mentions, (tsa, dhs, Markwayne mullin), and in its NMF sub-themes on immigration. (Figure 8)

Emotional register intensifies within the immigration reporting. The outrage gap increases on Immigration for four of the five outlets, indicating that headline sensationalism intensifies further on this contested topic. CNN sees the largest increase (4.50 to 6.69) and nearly catches Fox News at the top, though Fox still leads (7.03). AP is the lone exception: its outrage gap drops by 27% (4.27 to 3.12), the only outlet to decrease, consistent with AP's procedural framing. Within the emotion-by-emotion breakdown, the contrast between CNN-anger and Fox-fear reported in Section 5.2 holds sharply in the immigration topic. CNN's headline anger density reaches 7.71 per 100 tokens, while Fox News's headline fear reaches 7.63. Fox News also leads notably on headline trust (6.64). (Figure 9)

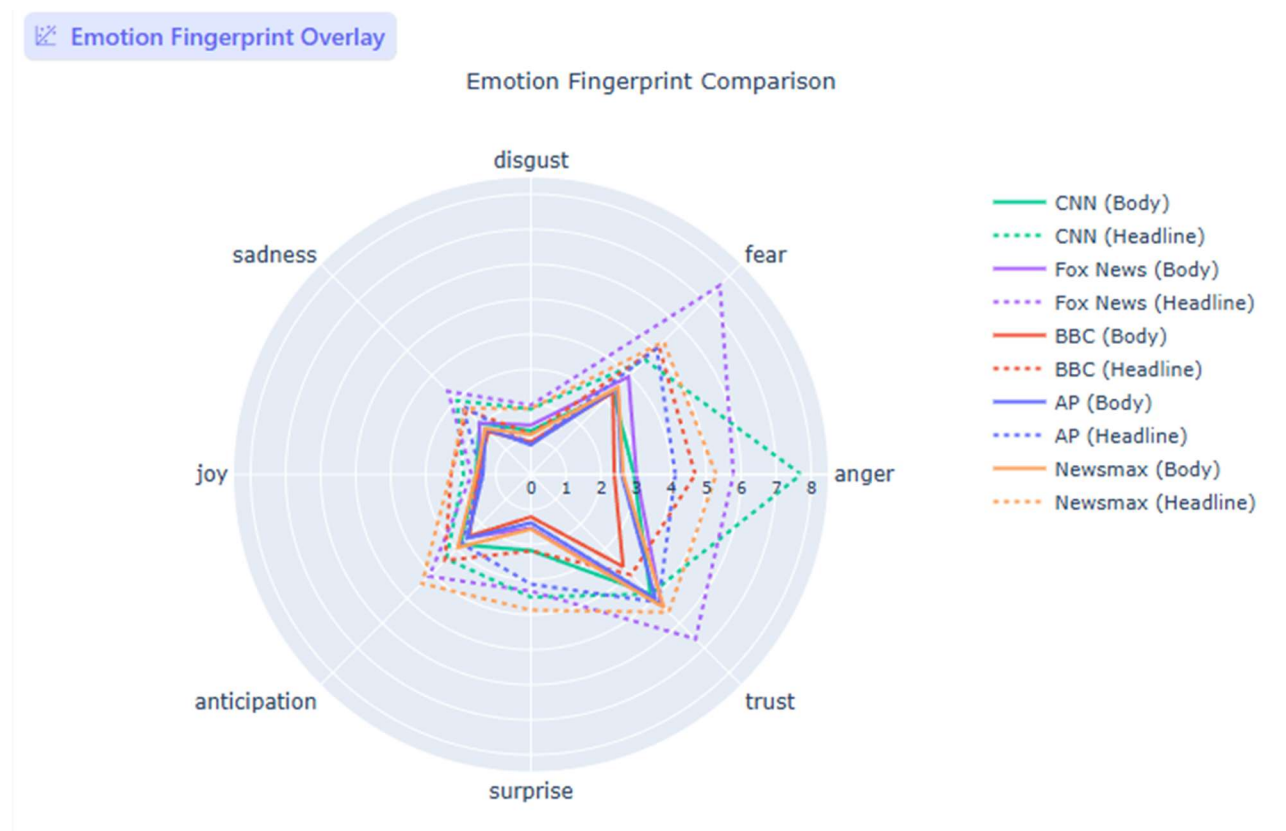


Figure 9, Radar chart of NRC emotional densities per outlet, split headline and body, filtered to Immigration topic articles

Moral framing shifts toward the binding foundations. Total moral-language density spikes between 26% and 41% across every outlet on Immigration, confirming the topic as structurally more moralizing than the corpus average. The spike concentrates in Loyalty and Authority, both of which rise sharply for every outlet. Loyalty roughly doubles for Fox and NewsMax, and Authority shows similar magnitude increases. Care/Harm shows a mixed pattern, as BBC and AP rise slightly while CNN and NewsMax fall. NewsMax leads on total moral density on Immigration, driven by the high Loyalty and Authority scores alongside its

high Fairness outlier. BBC again sits at the low end of total moral density, matching the corpus pattern, though the absolute distance from peer outlets narrows on this topic.

Considered together, the dimensions converge on a single characterization of Immigration coverage. Every outlet engages more intensively with immigration, using more moral and emotional language than they engage the corpus average. Loyalty and Authority spike across the corpus, and total moral-language density rises for every outlet. As the moral language rises uniformly across the outlets, the partisan outlets diverge sharply in emotional register. Fox News leans on fear, and CNN leans on anger, which are two emotional registers that political-psychology research has linked to distinct attitudinal effects. Anxiety and fear based coverage of immigration has been shown experimentally to drive opposition to immigration and support for restrictive policy (Brader, Valentino, & Suhay, 2008), a finding consistent with Fox News's emphasis on fear vocabulary in headline framing. Anger, by contrast, is the emotional register most associated with mobilization and oppositional engagement (Valentino et al., 2011), aligning with CNN's anger-heavy headline language as a possible appeal to push back against the enforcement actions the corpus covers. Whether outlets are stoking emotions their audiences already hold (Stroud, 2011) or actively shaping audience attitudes (DellaVigna & Kaplan, 2007), or more plausibly under the contemporary consensus, both operating in a mutually reinforcing loop (Levendusky, 2013), cannot be established from outlet-level content analysis alone. NewsMax does not follow Fox News into a fear-coded immigration register, as reporting on immigration is focused more on the issues in Washington, as opposed to enforcement policies. This can be seen in its notable mentions of the DHS, TSA, and Markwayne Mullin. This shows that editorial choice of which stories to run can also be a factor in how outlets portray certain topics.

This case study illustrates the value of analyzing outlet coverage at the topic level rather than only in aggregate. While the aggregate profile shows how an outlet leans in general, a topic-level view shows how that lean actually plays out on a specific news cycle. The topic level view shows where outlets across the political spectrum differ in reporting specific stories, and where outlets of similar editorial orientation produce visibly different framings. Both lenses can help news readers to understand media coverage fully.

7. Interactive Profile Explorer

The Immigration case study above demonstrates the value of reviewing a single topic across the outlets. To let readers inspect the same components for the other topics in the paper, and view the outlet profiles, an interactive web application was built. The explorer (hosted at <https://huggingface.co/spaces/Jack-O-Wood/news-outlet-profiles>) contains profiles for each outlet and a "Compare Outlets" tab for side-by-side viewing. On each page, a topic filter recomputes the entire profile within a single topic, allowing the reader to construct the same multi-dimensional view for AI, Climate, Economy, Immigration, or Iran. The interactive form is a prototype of a news profile tool that could help readers understand news coverage patterns and how outlets diverge in topic framing.

8. Discussion

Given the language-analysis approach taken here, the natural question is what these outlet profiles offer that existing media-bias ratings do not. The most widely used tools, such as AllSides, Media Bias/Fact Check, and Ad Fontes Media, characterize outlets primarily along a partisan left-right axis, viewed from a U.S. political perspective. Ad Fontes adds a second dimension, news value and reliability, on the y-axis, which provides an information-integrity angle but still collapses editorial behavior to two summary

scores. The language-analysis profiles built here do not replace these ratings, but supplement them with measurements of how outlets talk about the news. The profiles show what vocabulary they choose, which emotions they amplify, which moral foundations they invoke, and which entities they foreground.

These profiles also explicitly do not prescribe a partisan lean. Outlets are not going to escape their reputations of partisan positioning, and affective polarization has continued to deepen in the United States (Iyengar et al., 2019). The practical effect of a single bias score is often to filter readers to outlets with which they already agree. Depending on their label, readers tend either to embrace outlets as confirming their priors or dismiss them as "the other side." The multi-dimensional approach is intended to do something different. Instead of assigning a prior-confirming label, it gives them a description of the outlet's behavior that they can engage with. For example, the Ad Fontes Media Bias chart places Fox News and NewsMax in a similar position. While these two outlets do have a similar right leaning editorial direction, they cover the same topics in different manners, using different vocabulary and spotlighting different entities. In the immigration example, while Fox News used increased Fear language, NewsMax selected different stories for coverage. BBC offers a sharper version of the same point. Ad Fontes places BBC near the center of the U.S. political bias axis. However, the analysis here shows BBC operating structurally outside the U.S. political register entirely, with low moral-language density, UK-regional vocabulary, near-absent American national identity in named entities. These differences fall outside of a single left-right placement.

9. Limitations and Future Work

Methodologically, the dictionary-based methods used here (VADER, NRC EmoLex, MFD) capture the surface vocabulary of affect, emotion, and moral framing but do not model context, negation, sarcasm, or the relationship between words in a sentence. For example, a headline that reads "no improvement in the situation" registers a negation for which VADER cannot fully account. The MFD specifically conflates virtue and vice terms within each foundation, so an article praising care and one describing harm both score high on Care/Harm. Scores from these methods are best read as relative ordering across outlets within this corpus rather than as absolute or context-aware measures of affective or moral content. VADER was additionally developed and validated on social-media text, where intensifiers, capitalization, and punctuation conventions differ from professional journalism. While VADER is widely used in news-corpus research, the social-media training data is worth noting explicitly.

The corpus itself reflects a narrow temporal slice. The data covers just under a four-month timespan, from January 1 to April 22, 2026. The topical content is therefore weighted toward the specific news events of that period (Iran-Israel conflict and Strait of Hormuz tensions, DHS Shutdown, the AI infrastructure buildout, Virginia redistricting, and so on). A longer window or a corpus drawn from multiple periods would smooth this temporal bias and reveal which patterns are stable across news cycles versus which are artifacts of a particular moment. The sample within that window is also relatively small (2,334 articles total, with the smallest cell at 45 articles for NewsMax × Climate); larger per-cell sample sizes would tighten the estimates and enable more granular sub-topic breakdowns. The smaller sample reflects practical compute and time constraints rather than a methodological preference.

The set of outlets is constrained by data availability. MediaCloud coverage is uneven across U.S. news outlets. Several widely read sources such as The New York Times, The Washington Post, The Wall Street Journal, and major broadcast networks sit behind paywalls or restricted APIs that the pipeline used here

cannot access. The five outlets in this analysis are not a representative sample of U.S. news; they are the outlets that satisfied both ideological-spectrum and data-availability criteria. Expanding to a wider outlet set would require either institutional access or a different scraping strategy.

The analysis additionally pivoted away from event-matched sampling for practical reasons, which means cross-outlet comparisons within a topic are comparing each outlet's topical coverage rather than each outlet's coverage of the same specific stories. Differences in what outlets cover within a topic are therefore conflated with differences in how they frame what they cover. The selection-bias question understanding which stories each outlet chooses to cover at all can only partially fit under this paper's scope in the NMF sub-themes.

Finally, no human-annotation validation was performed. The original methodology plan included a human-annotation pilot in which annotators would label a subset of article pairs on dimensions like emotional intensity and framing direction, and the system's outputs would be checked against the human labels. This validation step was not completed in the time available. The analysis is therefore an internally consistent computational characterization of the corpus, but the question of whether the dimensional measurements would track human readers' perceptions of the same articles remains unanswered.

Beyond the limitations addressed above, three directions would extend this work most meaningfully. First, a selection-bias analysis would surface which stories each outlet chooses to cover at all. Second, longitudinal analysis would surface how outlet profiles shift around major events. Third, methodological upgrades from dictionary-based methods to transformer-based classifiers (which handle context and negation) would refine the affective and moral measurements. The interactive explorer presented in Section 7 is intended as a prototype rather than a finished product; further development for everyday news-reader use is a parallel line of work.

10. Conclusion

This paper showed that multi-dimensional profiles surface patterns single-axis bias ratings cannot capture: Fox News intensifies fear-coded vocabulary in headlines, CNN leans on anger as a consistent editorial register, NewsMax over-indexes on fairness-coded moral language, BBC operates structurally outside the U.S. political register, and outlets across the spectrum foreground entities representing the opposite of their stated lean.

The more durable contribution is methodological. No single dimension produced a defensible outlet profile on its own, but the dimensions converged. Fox News's headline distinctiveness surfaced independently in TF-IDF and NRC outrage scoring. BBC's outsider position appeared in moral foundations, entity selection, and regional vocabulary. That convergence across independent methods is what makes the resulting profiles interpretable as something more than artifacts of any single tool.

The interactive explorer is a prototype of what reader-facing media literacy could look like under this approach: not a label assigning a side, but a description of how an outlet talks. As affective polarization deepens and readers encounter more outlets through algorithmically curated feeds, tools that describe rather than judge may be the more useful intervention.

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